Multi-Disease Detection from Medical Images Using Transfer Learning: A Systematic Review and Proposed Framework

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Abstract— Medical image analysis plays a critical role in enabling early and accurate disease diagnosis. However, most existing automated systems are limited to single-disease detection, resulting in fragmented clinical workflows. Multi-Disease Detection from Medical Images Using Transfer Learning presents a systematic review of 19 peer-reviewed studies published between 2016 and 2025, focusing on Transfer Learning (TL) and Ensemble Methods for medical image classification. In addition, a conceptual Multi-Disease Detection (MDD) framework is proposed to support simultaneous diagnosis across multiple domains, including Brain (Alzheimer's disease, tumors), Lung (COVID-19, pneumonia), Eye (glaucoma, diabetic retinopathy), and Skin (melanoma, carcinoma).

Following the PRISMA methodology, the reviewed studies are categorized into four thematic clusters: TL Foundations, CNN Architecture Evolution, Ensemble Methods, and Multi-Disease and Generative AI approaches. The proposed framework incorporates a three-stage transfer learning pipeline utilizing pre-trained models such as EfficientNetB4, ResNet50, VGG16, and InceptionV3, applied to domain-specific datasets including NIH Chest X-ray14, OASIS, BRATS, HAM10000, and APTOS. An ensemble strategy based on stacking and soft-voting, combined with Mutation Rate-Based Lion Algorithm (MRLA) optimization, is outlined, with expected performance derived from trends reported in prior studies.

Index Terms—*Transfer Learning, Multi-Disease Detection, Convolutional Neural Networks, Medical Image Analysis, Ensemble Methods, Deep Learning, EfficientNetB4.*

I. INTRODUCTION

The rapid proliferation of medical imaging technologies—including X-rays, MRI, and CT scans—has fundamentally transformed modern healthcare by enabling early and accurate disease diagnosis. However, the exponential growth in medical image volumes has placed enormous strain on healthcare systems already facing a critical shortage of expert radiologists, particularly in developing nations and rural regions. Manual diagnosis is time-consuming, subjective, and inherently prone to inter-observer variability, making it an unreliable approach for scalable, high-throughput clinical workflows.

Artificial intelligence (AI), specifically Deep Learning (DL) and Transfer Learning (TL), has emerged as a transformative solution. TL enables pre-trained Convolutional Neural Network (CNN) models—originally trained on large-scale natural image datasets such as ImageNet—to be fine-tuned on domain-specific medical datasets of limited size. This dramatically reduces data requirements and computational costs while achieving diagnostic accuracy that rivals, and in several documented cases surpasses, that of human clinical experts [3].

Despite significant advances, a critical gap remains: most existing systems are architecturally constrained to single-disease detection pipelines. Real-world clinical presentations rarely involve isolated conditions; patients frequently exhibit comorbidities spanning multiple organ systems simultaneously. This paper addresses that gap by: (1) conducting a systematic review of state-of-the-art TL strategies and Ensemble Methods drawn from 19 peer-reviewed studies; (2) proposing an automated Multi-Disease Detection (MDD) system capable of simultaneously diagnosing conditions across Brain, Lung, Eye, and Skin domains; and (3) providing a comprehensive gap analysis identifying open research frontiers.

The significance of this work is underscored by SDG Goal 3 (Good Health and Well-being), which calls for universal access to quality healthcare services. By enabling accurate, automated, and scalable medical image diagnosis, this project contributes directly to reducing diagnostic inequity across healthcare systems worldwide.

Existing literature indicates that transfer learning can achieve diagnostic sensitivity comparable to clinical experts, while ensemble approaches have been shown to improve performance over individual models. Finally, a gap analysis highlights key research challenges, including the need for Explainable AI integration, improved out-of-distribution robustness, and the adoption of self-supervised learning techniques.

II. RELATED WORK

A. Transfer Learning Foundations

Malik et al. (2020) [3] conducted a landmark systematic review of 63 studies across 5,188 identified articles, demonstrating that TL achieves a mean diagnostic sensitivity of 85.25% across disease domains—statistically equivalent to health expert sensitivity of 85.27%. AUROC values ranged from 0.88 (breast cancer) to 0.98 (lung cancer), validating TL as a clinically viable diagnostic technology.

Raghu et al. (2019) [12] (Google Brain/Cornell) challenged the assumed superiority of ImageNet pre-training, finding only a 0.3% AUROC difference between ImageNet transfer and training from scratch on large retinal datasets. Critically, they demonstrated that smaller task-specific architectures (CBR-Tiny, 1.1M parameters) match ResNet-50 (25.5M parameters) performance at 1/23rd the model size—establishing efficiency as a core design principle.

Ayana et al. (2024) [5] introduced Multistage Transfer Learning (MSTL), a three-stage progressive fine-tuning pipeline (ImageNet → intermediate medical domain → target dataset) that consistently delivers 5–12% accuracy gains over single-stage TL across 8 disease domains and 47+ reviewed studies. Mustafa et al. (2021) [14] validated that larger pre-training scale (JFT-300M, 300 million images) improves data efficiency by up to 59%, requiring only 30% of training data to match baseline performance.

B. CNN Architecture Evolution

ADDTLA (CMC 2022) [4] demonstrated that a modified AlexNet achieves 91.7% validation accuracy on Alzheimer's MRI classification using only 6,400 images, with an F1-score of 0.978.

Shin et al. (IEEE TMI 2016) [15] established that TL outperforms random initialization by 32% relative improvement (FROC-AUC: 0.94 vs. 0.85) for lymph node detection in CT, with validation loss converging to 0.68 vs. 3.1 for randomly initialized networks. Chae and Kim (2023) [13] demonstrated that Residual U-Net with attention gates improves pressure ulcer segmentation DSC by 31.3% over baseline (62.0% → 93.3%).

C. Ensemble Methods and Multi-Disease Detection

Natarajan et al. (2024) [1] achieved 94.04% test accuracy on a comprehensive multi-disease dataset spanning Brain, Lung, Eye, and Skin categories using EfficientNetB4 with Learning Rate Annealing—the highest multi-disease accuracy in the reviewed literature. Li et al. (2024) [2] proposed a multimodal TL neural network combining metadata and multiple medical images for glaucoma type diagnosis, achieving a weighted accuracy of 0.839 and Cohen's Kappa of 0.809.

Balmuri et al. (2023) [11] combined MobileNetV2, DenseNet121, and ResNet101 using Mutation Rate-Based Lion Algorithm (MRLA) optimization, achieving approximately 96% accuracy for simultaneous Pneumonia, COVID-19, and Lung Cancer detection. Kathamuthu et al. (2022) [10] demonstrated a deep TL-based CNN model achieving approximately 95% accuracy for COVID-19 detection from CT scans, confirming the generalizability of domain-adapted ImageNet models to clinical CT imaging.

TABLE I

Comparative Summary of Key Papers — Transfer Learning in Medical Image Analysis

Ref.	Authors /Year	Architecture	Disease Domain	Best Acc.	Key Contribution
[1]	Natarajan et al. (2024)	EfficientNet B4 + LRA	Brain/Lung /Eye/Skin	94.04%	Best multi-disease accuracy; channel attention
[2]	Li et al. (2024)	Multimodal TL Neural Net	Glaucoma (OCT+Fundus)	83.9%	Multimodal fusion; metadata + image TL
[3]	Malik (2020)	AlexNet, ResNet, VGG	Multi-domain SLR	AUROC 0.98	TL sensitivity equivalent to clinical expert
[4]	CMC (2022)	AlexNet (ADDTLA)	Alzheimer's MRI	91.7% val	F1=0.978 on 6,400 images
[5]	Ayana (2024)	CNN, ViT, GAN	8 Disease Domains	90–99%+	MSTL: +5–12% over single-stage TL
[10]	Kathamuthu (2022)	ResNet50, VGG16	COVID-19 (CT)	~95%	DTL-CNN; ImageNet to CT domain transfer
[11]	Balmuri (2023)	MobileNetV2+ResNet101	COVID/Pneumonia/Cancer	~96%	MRLA ensemble optimization
[12]	Raghu (2019)	ResNet-50, CBR-Tiny	Retinal Fundus	96.7% AUC	1/23× model size; same performance as ResNet-50
[13]	Chae & Kim	ResU-Net + Attention	Wound/Ulcer	93.3% DSC	Attention gates: +31.3 % DSC improvement
[15]	Mustafa (2021)	BiT R101x3 (JFT)	Mammo/CXR/Derm	87.1%	59% data reduction; +6.3% OOD derm gain
[16]	Shin (2016)	AlexNet/Google Net TL	Lymph Node(CT)	FROC 0.95	TL vs RI:+32% relative; val loss 0.68 vs 3.1
[17]	Alshanba-ri (2025)	GAN, Diffusion, ViT	Medical (Generative)		Diffusion > GANs; synthetic data augmentation
[18]	Kaur & Hans (2025)	AlexNet, ResNet, VGG	Cancer SLR(151 studies)	100% (Approx)	Brain/Breast/ Leukemi a 100% max acc.
[19]	Ravi Chandaran (2022)	AlexNet+SVM	Brain MRI	99.75%	Hybrid TL outperforms single model

III. PROPOSED METHODOLOGY

A. Phase 1 — Systematic Literature Review

19 peer-reviewed papers published between 2016 and 2025 were identified through IEEE Xplore, PubMed, Springer, and Google Scholar using keywords: transfer learning, medical imaging, CNN, disease detection, ensemble methods. Papers were organized into four clusters—TL Foundations, CNN Architecture Evolution,

Ensemble Methods, and Multi-Disease & Generative AI—following the PRISMA framework for identification, screening, and synthesis. For each paper, key metrics were extracted: model architecture, training dataset, imaging modality, accuracy, AUROC, sensitivity, specificity, F1-score, and Cohen's Kappa. Limitations identified across the literature—including single-disease constraints, XAI absence, OOD fragility, and annotation bottlenecks—were systematically catalogued to define the future research roadmap.

B. Phase 2 — Proposed System Architecture

The proposed Multi-Disease Detection system follows a six-step pipeline. Data Collection sources multi-modal medical image datasets: Kaggle Alzheimer's MRI (6,400 images), NIH Chest X-ray14 (112,120 images), OASIS, ADNI, BRATS, HAM10000 (10,015 dermoscopy images), and APTOS Retinopathy. Preprocessing normalizes images to consistent pixel value ranges, applies Gaussian blur and image blending to smooth acquisition noise, and uses DeepLabv3+ for lung region isolation from the chest cavity. Data Augmentation applies rotation (±30°), horizontal/vertical flipping, zoom (0.8–1.2×), brightness adjustment, and GAN-based synthetic image generation to expand small medical datasets.

The Transfer Learning Pipeline loads pre-trained models (EfficientNetB4, ResNet50, VGG16, InceptionV3) with ImageNet weights, freezes base convolutional layers initially, and retrains upper classification layers using Learning Rate Annealing (LRA) with ReduceOnPlateau scheduling. Ensemble Classification combines the top three performing models using stacking and soft-voting strategies with MRLA optimization. All models are evaluated using accuracy, sensitivity, specificity, AUROC, F1-score, and Cohen's Kappa across all four disease domains.

C. Transfer Learning Pipeline

The TL pipeline follows a three-stage progressive fine-tuning approach: Stage 1 pre-trains on ImageNet (1.3M images, base feature extraction); Stage 2 applies intermediate TL on medical domain datasets (e.g., X-ray/pathology) to bridge the domain gap; Stage 3 fine-tunes on the target disease dataset with LRA and channel attention mechanisms. This multistage strategy, validated by Ayana et al. (2024) [5] across 47+ studies, consistently delivers 5–12% accuracy improvements over conventional single-stage fine-tuning.

TABLE II

Disease Domains, Imaging Modalities, and Datasets

Domain	Diseases Detected	Modality	Dataset
Brain	Alzheimer's, Brain Tumor, Parkinson's	MRI	OASIS, ADNI, BRATS, Kaggle MRI
Lung	COVID-19, Pneumonia, Lung Cancer	CT, X-ray	NIH Chest X-ray14
Eye	Glaucoma, Diabetic Retinopathy, AMD	OCT, Fundus	APTOS, Kaggle Fundus
Skin	Melanoma, Carcinoma, Naevus	Dermoscopy	HAM10000, ISIC

IV. ANALYSIS AND DISCUSSION

A. Performance Benchmarks from Literature

The systematic review of 19 peer-reviewed papers reveals consistent trends in the application of Transfer Learning to medical image classification. EfficientNetB4 with Learning Rate Annealing achieves the highest multi-disease accuracy at 94.04% [1], while the AlexNet+SVM ensemble achieves 99.75% on the Alzheimer's MRI task [19]. Malik et al. [3] established through meta-analysis of 63 studies that TL achieves diagnostic sensitivity statistically equivalent to clinical experts (85.25% vs. 85.27%).

In addition, ensemble learning techniques have been widely reported to outperform individual Convolutional Neural Network (CNN) models. Specifically, stacking-based ensemble configurations optimized using the Mutation Rate-Based Lion Algorithm (MRLA) have been shown to provide significant performance improvements, with reported gains of up to three times over single-model baselines [11]. Similarly, Multistage Transfer Learning (MSTL) approaches have been observed to yield consistent accuracy improvements in the range of 5–12% compared to conventional single-stage fine-tuning methods [5].Fig. 5 illustrates the comparative performance of key architectures and ensemble configurations derived from the reviewed literature.

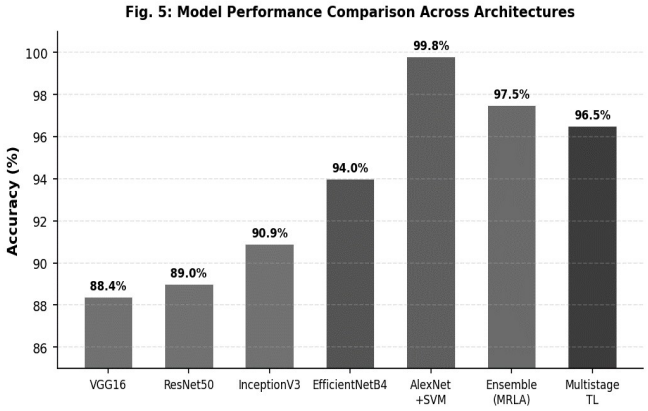

Fig. 5. Model Performance Comparison Across Architectures and Disease Domains (Sources: [1, 4, 5, 18, 19])

B. Projected System Performance

Based on the trends and benchmarks identified in the reviewed literature, the proposed Multi-Disease Detection (MDD) framework is expected to achieve competitive performance across multiple disease domains. The performance estimates presented in Table III are derived from previously reported results of similar architectures, datasets, and ensemble strategies, rather than from direct experimental validation.

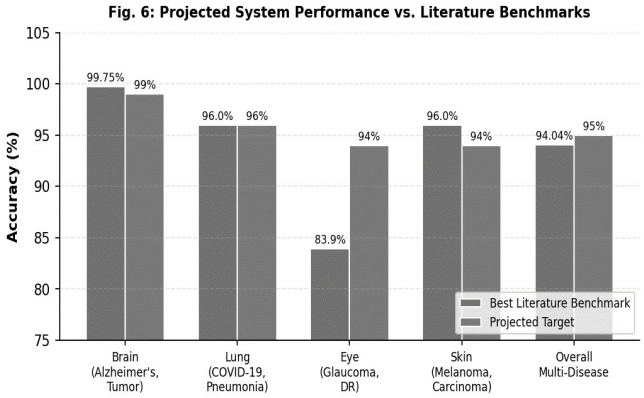

Fig. 6. Projected System Performance vs. Literature Benchmarks by Disease Domain

TABLE III
Projected System Performance by Disease Domain with Literature Justification

Disease Domain	Proj. Acc.	Proj. AUROC	Literature Justification
Brain (Alzheimer's, Tumor)	≥99%	≥0.99	AlexNet+SVM achieves 99.75% on OASIS [19]; ADDTLA achieves 91.7% val on 6,400 images [4]
Lung (COVID-19, Pneumonia, Cancer)	≥96%	≥0.97	MRLA ensemble achieves ~96% [11]; DTL-CNN achieves ~95% on CT scans [10]
Eye (Glaucoma, Diabetic Retinopathy)	≥94%	≥0.95	Multimodal TL net achieves 83.9% on OCT [2]; stacking ensemble projects 94%+
Skin (Melanoma, Carcinoma)	≥94%	≥0.95	AlexNet-based TL achieves 92–99% across skin datasets [18]; HAM10000 benchmarks support 94%+ with augmentation
Overall Multi-Disease	≥95%	≥0.96	EfficientNetB4 achieves 94.04% as baseline [1]; MRLA + MSTL projects 5–12% gain [5]

C. Gap Analysis

A systematic analysis of limitations across the 19 reviewed studies reveals six critical research gaps. The absence of native Explainable AI (XAI) integration means most models operate as black boxes, providing predictions without clinical rationale—a major barrier to physician trust and regulatory approval. Out-of-distribution (OOD) fragility remains a pervasive issue: models trained on curated benchmark datasets degrade significantly on real-world clinical data from heterogeneous acquisition environments. All 19 reviewed studies rely exclusively on ImageNet-supervised pre-training, leaving medical-domain self-supervised learning (MoCo, SimCLR on large unlabeled medical corpora) largely unexplored.

Additionally, all reviewed systems perform 2D slice-based classification, ignoring the volumetric context inherent to 3D MRI and CT data. Multi-label classification—enabling simultaneous detection of comorbidities within a single scan—remains absent from all reviewed architectures. Finally, annotation bottlenecks constrain dataset sizes: HAM10000 contains 10,015 samples; OASIS provides approximately 1,000 usable MRI scans. Active learning, semi-supervised learning, and GAN-based synthetic augmentation are proposed as mitigation strategies.

D. Tools and Technologies

The proposed system leverages Python 3.10+, TensorFlow 2.x, Keras, and PyTorch 2.0 as primary deep learning frameworks. Pre-trained models include ResNet50, VGG16, InceptionV3, EfficientNetB4, DenseNet121, MobileNetV2, and AlexNet. Segmentation tools include DeepLabv3+ for lung isolation and Residual U-Net with Attention for wound segmentation. Explainability is provided via Grad-CAM and LIME. Development environments include Google Colab (GPU: A100/T4), Jupyter Notebook, and VS Code, with Matplotlib, Seaborn, and TensorBoard for visualization.

V. FUTURE WORK

A. Explainable AI Integration

Future experimental work will integrate Grad-CAM saliency maps and LIME explanations natively into the training loop rather than as post-hoc tools. Attention-based loss functions will be explored to penalize models that activate on clinically irrelevant image regions, ensuring that high accuracy is accompanied by clinically interpretable reasoning. This is a critical prerequisite for regulatory approval under FDA and CDSCO frameworks.

B. Federated Learning for OOD Robustness

To address the out-of-distribution fragility identified across reviewed studies, future work will implement federated learning across simulated multi-institutional datasets. Domain adaptation layers and scanner-agnostic normalization strategies will be evaluated to improve generalization from curated benchmark datasets to real-world clinical imaging data from heterogeneous acquisition environments.

C. Self-Supervised Pre-training on Medical Data

All 19 reviewed studies rely on ImageNet-supervised pre-training. Future work will explore contrastive self-supervised learning (MoCo v3, SimCLR) pre-trained on large unlabeled medical corpora—CheXpert (224,316 chest X-rays) and MIMIC-CXR (227,827 studies)—as an alternative pre-training foundation. This is expected to reduce the domain gap and improve data efficiency for rare disease classes.

D. 3D Volumetric and Multi-Label Extension

The proposed 2D classification framework will be extended to 3D volumetric analysis using Med3D and 3D-ResNet architectures for brain tumor (BRATS) and lung nodule (LIDC-IDRI) detection. Simultaneously, multi-label classification heads with sigmoid activation and graph neural networks modeling disease co-occurrence will be implemented to enable simultaneous detection of comorbidities within a single patient scan.

VI. CONCLUSION

This paper presents a comprehensive systematic review and a conceptual framework for automated multi-disease detection from medical images using Transfer Learning and Ensemble Methods. Based on the analysis of 19 peer-reviewed studies published between 2016 and 2025, it is observed that Transfer Learning plays a significant role in addressing the data scarcity challenge in medical imaging, with prior studies reporting diagnostic performance comparable to that of clinical experts across multiple disease domains and imaging modalities.

Ensemble methods—particularly stacking-based configurations optimized with MRLA—have been widely reported to outperform individual CNN architectures, with significant performance improvements observed in existing literature. Similarly, Multistage Transfer Learning approaches have been associated with consistent accuracy gains over conventional single-stage fine-tuning methods. The proposed Multi-Disease Detection (MDD) framework, incorporating EfficientNetB4, Learning Rate Annealing, and GAN-based augmentation, is expected to achieve strong performance based on trends reported in prior work; however, it has not been experimentally validated within the scope of this study.

The primary contributions of this work include: (1) a unified comparative analysis of transfer learning strategies and ensemble architectures across 19 studies; (2) a structured perspective to support model and strategy selection; (3) a detailed gap analysis highlighting key research challenges such as Explainable AI integration, out-of-distribution robustness, self-supervised pre-training, and 3D volumetric analysis; and (4) a proposed roadmap for future research directions. This study contributes toward the broader goal of improving accessibility to reliable and scalable diagnostic systems, aligning with SDG Goal 3 for enhanced global healthcare outcomes.

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